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The Use of Next-generation Sequencing in the Diagnosis of Rare Inherited Anaemias: A Joint BSH/EHA Good Practice Paper

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METHODOLOGY

The British Society for Haematology (BSH) produces Good Practice Papers to recommend good practice in areas where there is a limited evidence base but for which a degree of consensus or uniformity is likely to be beneficial to patient care. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) nomenclature was used to evaluate levels of evidence and to assess the strength of recommendations. The GRADE criteria can be found at http://www.gradeworkinggroup.org.

This Good Practice Paper was produced as a collaboration with the European Hematology Association (EHA) compiled according to the BSH process at http://scanmail.trustwave.com/?c=8248&d=68DV3b1jbPPsVn. This guideline group included UK-based medical experts representing the BSH and members of the EHA Red Cell and Iron Scientific Working Group (SWG).

Literature review details

MEDLINE, EMBASE and PubMED were searched systematically for publications in English from 2000 to 2019 using the following key words. 'NGS' and 'next-generation sequencing' or 'high throughput sequencing' AND 'haemolytic anaemia' or 'DBA' or 'Diamond Blackfan anaemia' or 'CDA' or 'congenital dyserythropoietic anaemia' or 'sideroblastic anaemia' or 'HS' or 'hereditary spherocytosis' or 'red cell membrane disorders' or 'red cell enzyme disorders' or 'PK deficiency' or 'PKD'. References from relevant publications were also searched. Conference abstracts were included if deemed to be of particular relevance.

Review of the manuscript

Review of the manuscript was performed by the BSH Guidelines Committee General Haematology Task Force, the BSH Guidelines Committee and the General Haematology sounding board of the BSH. It was also on the members section of the BSH website for comment. It has also been reviewed by members of the EHA Red Cell and Iron SWG and the EHA Guidelines Executive Committee.

INTRODUCTION

The use of next-generation sequencing (NGS) in the diagnosis of rare inherited anaemias is increasingly common, as evidenced by a growing number of publications describing its clinical utility.[1–6] Excluding disorders of globin synthesis, rare anaemias include Diamond-Blackfan anaemia (DBA), congenital dyserythropoietic anaemias (CDA), congenital sideroblastic anaemias (CSA), and disorders of red cell membrane and

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enzymes. Other forms of genetic anaemias can also be considered while establishing NGS panels, in particular genetic syndromes, where anaemia comprises one of the constellation of symptoms.

Table 1 briefly summarises the key aspects of these conditions.

The advantages of using NGS over single-gene testing, in addition to the cost effectiveness, is that clinical and laboratory features are often not specific for a particular condition, and a large number of large candidate genes might need to be analysed before making a diagnosis. A proportion of the patients also present with overlapping phenotypes, and it has been shown that in 10%–40% of cases, there is a degree of misdiagnosis or no diagnosis when this is based purely on phenotype and traditional non-NGS testing.[1,6] This can result in incorrect or inadequate treatment, causing anxiety and adversely affecting quality of life and potentially cost.

The term 'NGS' refers to all types of high-throughput sequencing, and for the purposes of this good practice guideline will include targeted resequencing (t-NGS), whole exome sequencing (WES) and whole genome sequencing (WGS). Table 2 shows the advantages and disadvantages of each type of NGS. A detailed description of NGS techniques is beyond the scope of this paper; however, this is summarised in Figure 1. In t-NGS, only the genes selected are sequenced, whereas in WES ~30 000 genes are sequenced and in WGS all genes and intergenic regions are sequenced. However, in WES and WGS, the coding sequences of only a subset of genes are analysed, what is frequently referred to as a 'virtual panel'. In addition, coverage of genes is best in WGS where no DNA amplification step is required. Large duplications and deletions, involving one or more whole genes, known as copy number variants (CNVs), are more difficult to identify, but can be detected using appropriate analysis, particularly using WGS, but also WES and targeted resequencing.

It is important to note that, depending on the size of the panel, a number of variants will always be identified after all of the filters are applied, even in normal individuals. This number will depend on the number of genes, the inherent polymorphic potential of the gene and the ethnic origin of the individual tested. All variants identified post-filtering need to be assessed against strict criteria to determine their pathogenicity, based on the guidelines of American College of Medical Genetics (ACMG).[7] It is good practice to assess all variants even after a pathogenic variant has been found, to help with interpretation if this variant is identified again in the future.

Excellent comprehensive guidelines exist for the preparation of samples and the quality control that should be followed.[8] Likewise, the ACMG and Association of Clinical Genomic Science (ACGS) guidelines detail the interpretation of variants, and all laboratories should follow these criteria to determine pathogenicity of all variants identified.[7,9] The ACGS guidelines are less stringent in their assessment of evidence for pathogenicity. The ACMG system therefore scores more variants as variants of uncertain/unknown significance (VUS)/class 3 than the ACGS guidelines, increasing specificity at the expense of sensitivity.

The purpose of this paper is to give guidance on the uses of NGS that are specific to the diagnosis of rare inherited anaemias. This may be useful to laboratories wanting to set up NGS or for ones that have set this up for research and are planning to use it for clinical diagnosis. The type of NGS used, the conditions for which it can be used and the timing of it in the diagnostic pathway will partly depend on each country's healthcare system and funding arrangements. However, we aim to issue general guidance. Most of the guidance below is best suited to t-NGS as this is currently most commonly used, but the principles are equally applicable to the other technologies.

This good practice paper will address the following questions:

1. When is NGS necessary or of additional value in the diagnosis of rare anaemias?
2. At which point in the diagnostic pathway should NGS be used?
3. What are the important considerations in choosing the most appropriate NGS method and which quality criteria must be met?
4. What criteria should be used for reporting NGS variants identified?
5. How should variants identified be stored and shared between laboratories?
6. What criteria are essential for a laboratory to be able to offer clinical-grade NGS?

Table 1.

Key Aspects of the Rare Anaemias Not Due to Disorders of Haemoglobin Synthesis

	DBA	CDA	Sideroblastic Anaemia	Red Cell Membrane/Cation Leaking and Enzyme Disorders
Age at presentation	Usually 2–3 mo of age or <first year of life	Usually child/young adult	Usually child/young adult	Foetal/neonate/child/young adult
Associated features	Craniofacial Skeletal Cardiac Urogenital tract	Distal limb Iron overload	Ring sideroblasts on bone marrow aspiration	Jaundice Hepatosplenomegaly Gallstones Iron overload Progressive myopathy and neurocognitive impairment[a] Lymphoedema[b]
Severity	Moderate to severe	Usually mild to moderate	Mild to severe	Mild to severe
Treatment	Corticosteroids Transfusions and chelation BMT	Interferon Transfusions and chelation Often none needed	Transfusions and chelation Often none needed	Often none needed Splenectomy Transfusions and chelation New agents
Genetics	Autosomal dominant (45%) or de novo (other inheritance for DBA-like disease) Ribosomal proteins or other genes affecting ribosome biogenesis (other genes for DBA-like disease)	Autosomal recessive or dominant, X-linked Vesicle trafficking, heterochromatin assembly, nuclear proteins, transcription factors	X-linked; autosomal recessive Haem synthesis	Autosomal dominant or recessive; X- linked RBC membrane cytoskeleton, RBC transporters and RBC enzymes

[a]Associated with some rare enzymopathies or rare form of glucose transporter type 1 (GLUT1) variants.

[b]Associated with some severe form of hereditary stomatocytosis.

CDA = congenital dyserythropoietic anaemia; DBA = Diamond-Blackfan Anaemia; RBC = red blood cell.

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Table 2.

Comparison of Different Types of Next-Generation Sequencing

	t-NGS	WES	WGS
Target of sequencing; size (base pairs [bp])	Exons of 20–200 genes with some intron/exon boundaries for splice site mutations; 500 000 bp	The 'exome' ~30 000 exons of known coding genes (~1.5% of genome but 80%–90% of known disease-causing mutations) with some intron/exon boundaries for splice mutations; 2×10^7 bp	The whole genome (coding and noncoding space) 3×10^9 bp
Method	Capture of chosen exons, amplification steps and sequencing or amplification of chosen exons and sequencing	Capture all the exons, amplification step and sequencing	DNA is fragmented randomly, ligation of adaptors and direct sequencing (no capture or amplification)
Advantages	Cost, relative ease of interpretation, few unsolicited findings, more challenging to identify CNVs	Cost lower than WGS	Entire genome interrogated including non-coding region; more potential to identify CNVs. Can add genes to virtual panel. Relatively even coverage
Disadvantages	Will only identify mutations in targeted regions, coverage is often uneven, so mutations may be missed. Harder to detect some CNVs	Interpretation can be challenging, high chance of unsolicited findings, will only find mutations in coding regions, coverage is often uneven, may not detect CNVs. Ethical issues of incidental findings in genes that predispose to serious illness	Interpretation challenging unless there is a trio, non-coding region cannot easily be interpreted. Ethical issues of incidental findings in genes that predispose to serious illness Cost

CNV = copy number variant; t-NGS = targeted resequencing panels; WES = whole exome sequencing; WGS = whole genome sequencing.

Question 1: When is NGS necessary or of additional value in the diagnosis of rare anaemias?

Most current NGS approaches include the genes involved in the pathology of DBA, CDA, CSA, and disorders of red cell membranes and enzymes.[1,6] The globin genes are frequently but not always included. First, much of globin gene testing required for pre- and neonatal diagnosis requires a rapid turnaround time and analysis of a small number of genes, making it unwieldy and unnecessary to be testing all of the genes on a panel. In most cases, the clinical and laboratory presentation is clear and only a minority have a differential diagnosis of other haemolytic anaemias. Second, these are regions of very high-sequence homology, potentially resulting in poor specificity and high levels of artefacts and false-positive results on NGS testing, depending on the specific technology selected. In addition, many of the pathogenic genetic abnormalities leading to haemoglobinopathies are CNVs (insertions or deletions), which can be more difficult to detect by t-NGS. In particular, some common alpha globin variants such as the 3.7 kb deletion and triplicated alpha globin gene, are especially challenging as the breakpoint sequences are not unique. However, robust validation of the panel can ensure the reliable detection of most globin variants and some panels have been designed specifically to detect CNVs in globin genes, enabling the option of using NGS for haemoglobinopathy diagnosis.

There are circumstances when globin gene sequencing is of particular importance, including in the assessment of microcytic or haemolytic anaemias. Haemoglobin subtype analysis, including the quantitation of haemoglobin A_2, can identify or exclude most globin gene variants, but does not reliably identify many cases of unstable haemoglobin, dominant thalassaemia,[10,11] or individuals with beta thalassaemia intermedia resulting from heterozygous of beta thalassaemia in the presence of triplicated alpha gene.[12] In the case of unstable haemoglobins, the patients may have a mild to severe haemolytic anaemia, including transfusion dependence.[13,14] The unstable haemoglobin is often not detectable using haemoglobin analysis, and the presence of transfused blood also makes phenotypic diagnosis more difficult, particularly if started neonatally. Globin gene variants are the commonest cause of inherited anaemia, and all patients should be formally assessed for their presence, using a combination of haemoglobin analysis and specific genetic tests for suspected variants, and by inclusion on NGS panels, depending on local practice. Particular consideration needs to be given to excluding CNVs of the alpha globin genes, which may require specific assays using a gap polymerase chain reaction (Gap-PCR) or multiplex ligation-dependent probe amplification (MLPA).

Devising a list of conditions and genes to include in the t-NGS or virtual panel

The number of genes to include in a panel must balance inclusivity, to reduce false-negative rates, with increasing workload from needing to review and critically assess a large number of variants.

For any laboratories wishing to set up t-NGS for rare inherited anaemias, Suppl. Table S1 contains our suggested list of genes. Any published list is rapidly out of date as new evidence accumulates. However, the majority of known genes will be valid for some time. In England, genetic testing has been harmonised nationally and all the genes on each panel offered are available on PanelApp: https://panelapp.genomicsengland.co.uk/. This list of genes has been determined and curated by specialists in the field and is updated yearly to ensure that newly published genes are included.

It is worth considering if there are conditions in which NGS is of no added value and whether the reluctance to use NGS in some cases is purely due to its cost. There are rare anaemias that are often straightforward to diagnose without recourse to DNA analysis, for example, hereditary spherocytosis (HS). Nevertheless, for such cases the advantage of carrying out molecular analysis is that it facilitates genetic counselling. This can be especially helpful in some HS cases without a clear family history, to distinguish between recessive inheritance and a de novo variant. Conversely, laboratory tests for HS reach a sensitivity/specificity of >98%/90%, which is higher than for t-NGS. Although these are often mild conditions, they can result in significant morbidity including foetal anaemia, kernicterus and transfusion dependence, and genetic counselling is useful, particularly in families who wish to avoid further affected pregnancies.

It is particularly important to be certain of the precise diagnosis before performing splenectomy for presumed HS, to avoid ill-advised splenectomy in dehydrated hereditary stomatocytosis as this procedure is accompanied by a greatly increased risk of thromboembolic disease.[15] Phenotypically these conditions can be very similar unless some assessment of red cell hydration is performed, such as osmotic gradient ektacytometry or osmotic fragility measurement. In general, genetic diagnosis should be confirmed before recommending splenectomy in HS, and this will typically involve analysis using an NGS panel. Additionally, documenting genetic variants will eventually lead to some genotype–phenotype correlations.[16,17] This is the case with pyruvate kinase (PK) deficiency, where response to the new drug AG-348 depends on whether the mutations are missense or not.[18]

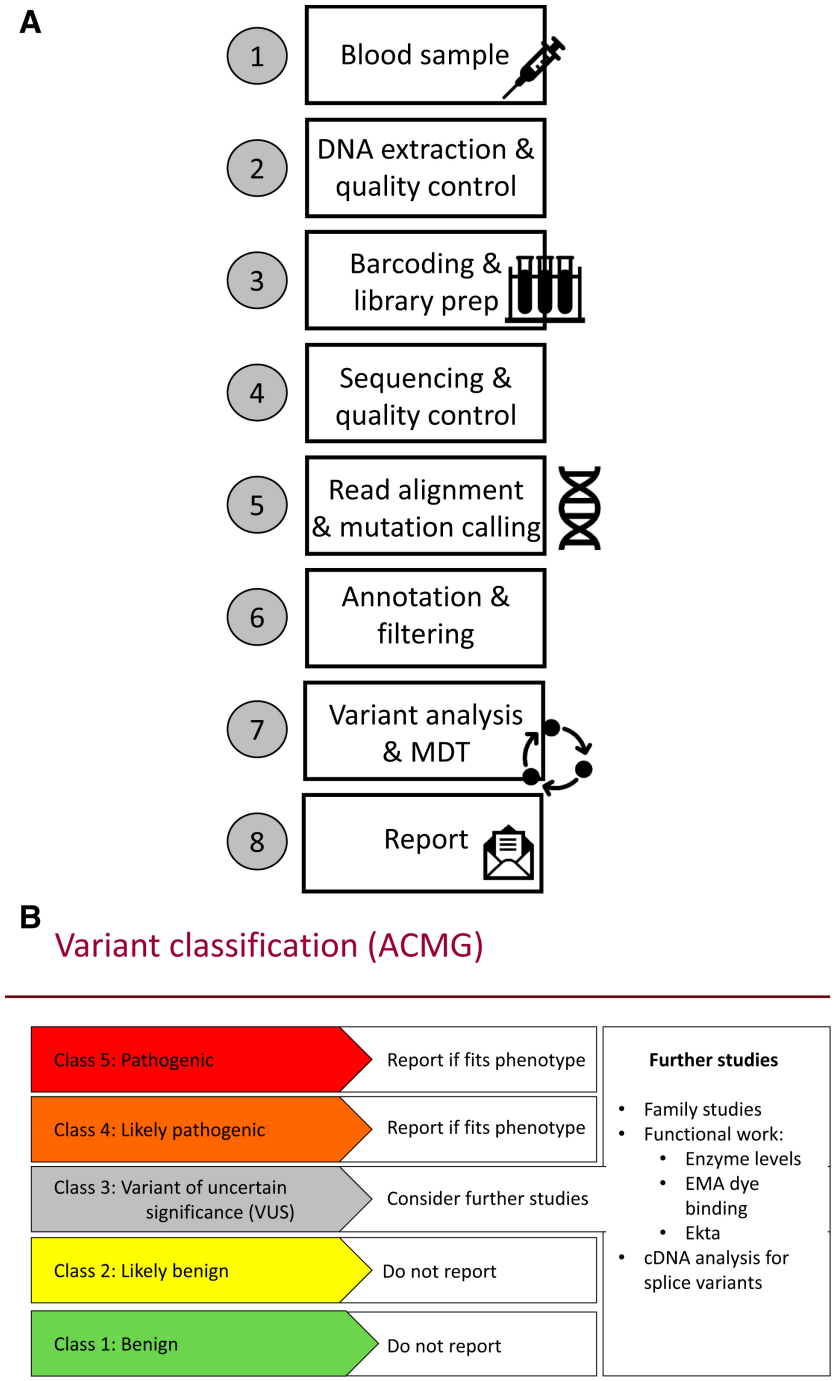

Figure 1. (A) Cartoon of the process of creating an NGS report from arrival of sample in the laboratory. (Usually includes clinical scientists and clinicians.) (B) American College of Medical Genetics variant classification, with examples of further studies that can be carried out to determine the pathogenicity of class 3 variants of uncertain significance. This includes family studies to investigate segregation, as well as functional assays such as red cell enzyme activities, EMA dye binding for hereditary spherocytosis, and osmotic gradient ektacytometry (Osmoscan), which investigates red cell deformability for membrane disorders. This list is not exhaustive and includes other functional assays (eg, electron microscopy for CDA, ribosomal profiling, or northern blots for DBA); EMA dye binding. CDA = congenital dyserythropoietic anaemias; DBA = Diamond-Blackfan anaemia; Ekta = ektacytometry; EMA = eosin-5'-maleimide; MDT = multidisciplinary team; NGS = next-generation sequencing.

For some conditions, NGS is far superior to Sanger sequencing of specific genes, due to the phenotypic variability and the unreliability of phenotypic tests such as enzyme assays for rare enzymopathies, making it difficult to target genes precisely, particularly when the patient is transfusion dependent. Because of frequent misdiagnosis of 'dyserythropoiesis' in some haemolytic anaemias,[1,6] genetic analysis should always be used to confirm a 'CDA'.

One condition where genetic analysis is particularly useful is dehydrated hereditary stomatocytosis (xerocytosis) due to autosomal dominant mutations in the gene Piezo-type mechanosensitive ion channel component 1 (*PIEZO1*), a mechanosensitive calcium channel. Patients with this condition are probably at high risk of developing post-splenectomy thrombosis and splenectomy in these cases is generally contraindicated.[15,19] This condition is difficult to diagnose and can be associated with

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only occasional stomatocytes on the blood film; genetic diagnosis should usually be performed before splenectomy when there is a possibility that the diagnosis could be dehydrated hereditary stomatocytosis; this will include most cases of presumed HS.

Finally, NGS-based genetic testing is useful for the identification of complex modes of inheritance that are recognised to account for at least 4% of diagnosed Mendelian conditions.[20]

Recommendations

- NGS should only be used in cases where acquired causes are thought to be very unlikely (IA)
- Appropriate consent should be obtained (IA)
- Globin gene abnormalities should be considered and investigated appropriately before NGS is carried out, including haemoglobin analysis and sequencing of individual globin genes, depending on the genetic distribution that is already known in the local population. Specific consideration should be given to globin gene CNVs, with use of Gap-PCR and MLPA as appropriate (IIB)
- Conditions that should be tested on the panel include DBA, CDA, CSA, suspected red cell enzyme deficiencies and red cell membrane disorders (IIB)
- Genetic analysis should be used to confirm conditions when there is diagnostic uncertainty (IIB)
- Genetic analysis should be performed before undertaking splenectomy for inherited haemolytic anaemias or other irreversible procedures such as bone marrow transplantation, where the genetic variant should be excluded from a potential stem cell sibling donor (IIB)

Question 2: At which point in the diagnostic pathway should NGS be used?

The use of NGS will partly depend on each country or hospital system's technical and reimbursement characteristics. The traditional investigative pathway is to take a history and examination, full blood count, reticulocyte count, and haemolytic markers, before selecting specialised tests (enzyme assays, osmotic gradient ektacytometry, eosin-5-maleimide [EMA] test, erythrocyte adenosine deaminase [eADA], etc.). In some cases, this may lead to a bone marrow biopsy or aspiration, with genetic analysis being kept at the end of the pathway. In other places, genetic analysis may occur much earlier in the pathway.[21] The advantages are that this may lead to a more rapid diagnosis, may be cost effective in reducing delay in diagnosis (at the expense of a higher cost upfront) and may (in some conditions) preclude the need for a bone marrow biopsy. Figure 2 shows examples of aspects of the history and examination that should be sought when evaluating the patient, as well as standard blood tests. The requirement for specialised tests, bone marrow aspiration and biopsy, and genetic analysis and the order in which they are requested, will differ between services, but in time, genetic analysis is likely to be carried out earlier in the pathway, with specialised functional analysis used to confirm the genetic diagnosis.

Recommendations

- NGS should primarily be used once the phenotype has been characterised. In particular, it should be established whether the patient has haemolysis, ineffective erythropoiesis, dyserythropoiesis, or bone marrow failure, as this may direct the analysis of the variants identified (IC)
- Clinical-grade NGS should ensure that variants are reported with reference to the phenotype of the patient (a sample request form detailing minimal phenotypic information can be found in Suppl. Figure S1) (IC)
- If further investigations are required to confirm the diagnosis (eg, family studies, RNA studies, specialist haematological tests directed by the variant identified), these can be recommended on the genetics report (IC)

Question 3: What are the important considerations in choosing the most appropriate NGS method and which quality criteria must be met?

Most panels are currently carried out as t-NGS, although some diagnostic laboratories carry out target enrichment across thousands of regions, then analyse the variants among genes that have been grouped together into virtual panels. As some countries move towards conducting all genetic analysis in the form of WGS, virtual panels will be increasingly used. The choice of using t-NGS over virtual panels is mostly due to availability, cost and turnaround time. Although cost-per-base may be lower for WGS, this requires a capital investment beyond the scope of most diagnostic laboratories. However, a major disadvantage of using t-NGS is that if any new genes are found to be associated with a known phenotype, adding a gene to the panel requires complete redesign and revalidation. This time-consuming and expensive process limits updating t-NGS panels to about once a year. WGS is also better suited for determination of CNVs, a common genetic cause of a number of inherited anaemias, with alpha globin gene deletions remaining a particular challenge for all technologies. New bioinformatic protocols to improve CNV assessment from targeted panels are improving their detection across modalities. Bait capture and unique molecular indexed amplicon methods may be combined with bioinformatic algorithms to determine the breakpoint mapping from short reads.[22] As the selected method will depend on many factors, it is critical that a laboratory is aware of the limitations of the technique, and that additional steps are taken to either overcome some of these limitations (eg, gap-filling by Sanger Sequencing) or that the report produced is explicitly clear on the limitations of the analysis. This may require suggesting alternative methods (eg, MLPA) to address CNVs that may not be detected reliably by t-NGS.

The availability of complementary diagnostic tools such as erythrocyte morphology, red cell and reticulocyte indices, EMA dye binding or osmotic gradient ektacytometry for red cell membrane disorders, may allow a phenotypic confirmation of the diagnosis in the absence of a definitive genetic diagnosis.

Recommendations

- The NGS method should be chosen based on local resources and required turnaround time (IC)
- Depending on the method chosen, the laboratory should be aware of the limitations and either reduce these (MLPA, gap-fill) or make it clear in the report what has not been tested (IC)

Question 4: What criteria should be used for reporting NGS variants identified?

Once variants have been identified and graded for pathogenicity, a multidisciplinary team meeting (MDT) is carried out, where variants are discussed in the context of the clinical presentation and a final report is written. In cases of an established pathogenic variant that fits with the phenotype, a report can be issued by the clinical scientists in the absence of an MDT meeting.

The ACMG guidelines must be followed for pathogenicity of single-nucleotide variants (5 classes)—pathogenic (class 5) and likely pathogenic (class 4) variants related to the clinical suspicion should be included in the report.

Recommendations

Variant types to include in the final report (IIB):

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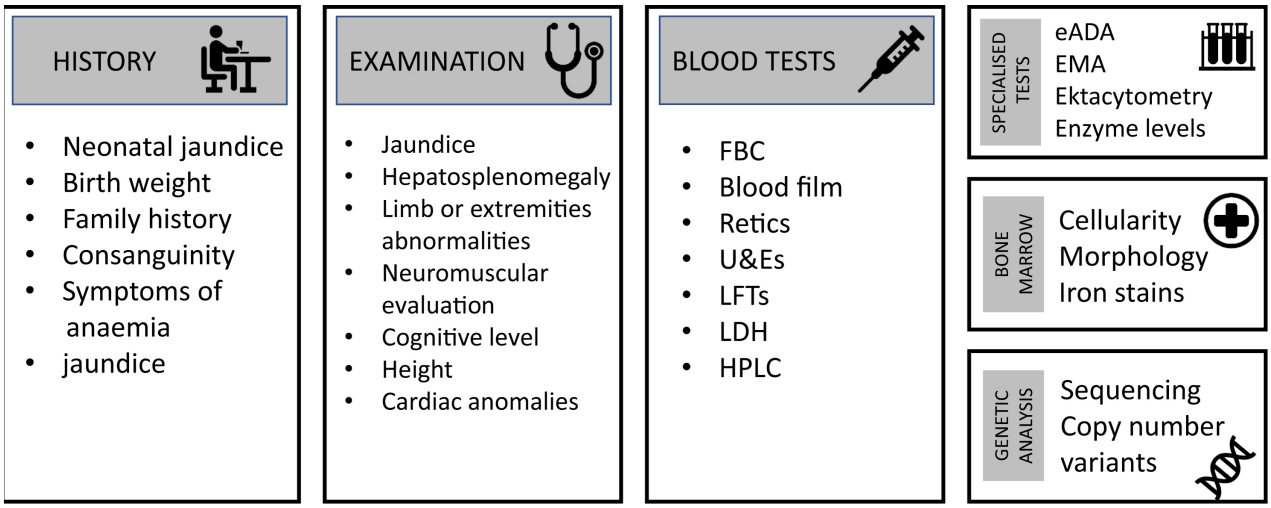

Figure 2. Clinical and laboratory assessment of the patient with a suspected diagnosis of inherited anaemia. These are indicative only and not exhaustive. eADA, erythrocyte adenosine deaminase; EMA, eosin-5′-maleimide test; FBC, full blood count; HPLC, high performance liquid chromatography; LDH, lactate dehydrogenase; LFTs, liver function tests; retics, reticulocytes; U&Es, urea and electrolytes.

1. Pathogenic/likely pathogenic variants related to the clinical suspicion

Variants to which a pathogenic role can be attributed with certainty, including:

- known variants in genes already associated with phenotype/disease
- novel variants in genes already associated with the phenotype/disease that have a clear causative role (eg, loss-of-function of a known gene that is associated with disease with a mechanism of haploinsufficiency), and fits with the pattern of inheritance, if available.

2. Pathogenic variants unrelated to the clinical suspicion

Variants with a well-known pathogenic role not related to clinical suspicion, including:

- causative variants in genes already associated with a phenotype but different from the suspected disease (*reverse phenotyping*)
- incidental findings (eg, carrier state for other condition), which should be reported only if consent explicitly signed for this as per the ACMG guidelines.

3. Variants with unknown clinical and functional role (*VUSs*) that could provide a diagnosis pending further investigation or evidence

These variants can be identified in:

- genes related to the suspected phenotype, which can be included in the final report. However, it should be made clear that the variant is a VUS and that without functional or family studies one cannot be sure that this variant is involved in the pathogenicity of the condition.
- genes not related to the clinical suspicion, which should not be included in the report.

In general, it is not recommended that intronic/splice (non-canonical) and 5′ and 3′ variants are reported unless substantial functional data is available. Some laboratories may report a recessive disorder where one pathogenic mutation (classes 4 or 5) has been found together with a VUS. Family studies are strongly recommended, and the report must make clear that there is no definite pathogenicity associated with the second variant.

This also includes circumstances where 2 very rare VUSs are identified in a gene(s) implicated in the phenotype, and family studies indicate they are in *trans* and functional data supports this gene as being causative. Variants of uncertain significance or variants that would suggest a novel complex mode of inheritance can form the basis of research studies, with the caveat that this almost universally requires a different form of consent to that obtained for diagnostic testing.

Question 5: How should variants be stored and shared between laboratories?

The sharing of variants between laboratories plays a very important role in ensuring high-quality data, high-diagnostic rates, and cost efficiency. However, this is often much more difficult to achieve than might be imagined, with issues such as data storage and the practicalities of sharing variants being significant obstacles.

One of the prerequisites for variant sharing is that participating laboratories use the same system for variant classification. Sharing of variants is difficult because ideally the information to be shared includes the clinical phenotype, how the pathogenicity was assessed including individual components of the overall score, and knowledge of the other variants found in the same patient. A potentially pathogenic variant where a different definitive genetic cause has also been found in the patient means the first one is less likely to be pathogenic.

However, the more variants are shared, the more identifiable the data are, raising the possibility that individuals may be identified according to specific haplotypes. Other obstacles to routine variant sharing across laboratories include practical technical reasons (not everyone shares and stores data in the same way) and time (the need to keep the database up to date and curated, with someone to take responsibility for any discrepancies).

Variant sharing is also predicated upon using the same nomenclature (eg, Human Genome Variation Society [HGVS]) and reporting against the same transcripts. In cases of multiple transcripts, the specific 'disease transcript' must be used, but this is not always known. Laboratories should make reasonable efforts to ensure that the transcript they are using is expressed in erythroid cells. LRG (Locus Reference Genomic) may be useful in this assessment: https://www.lrg-sequence.org.

Recommendation

- Laboratories should share variants with other laboratories analysing the same genes (IIC)
- Laboratories should ensure they are using commonly used transcripts which have been shown to be expressed in erythroid cells (IIC)

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- Ethical and legal issues in sharing variants between laboratories, often located in different countries should be clearly reported and discussed (IIC)

Question 6: What criteria are essential for a laboratory to be able to offer clinical-grade NGS?

For a laboratory to offer clinical-grade NGS, a number of parameters must be met, which relate to the laboratory itself, the panel design, the analytical pathway and the report.

Patient consent

This will depend on the legal framework of each country. However, some general guidance can be issued. Each laboratory should have a clear, written policy on consent, which is available to referring clinicians, and takes into account the following considerations:

- the risk of identifying non-paternity
- incidental findings (this will depend on the gene content of the panel, the way it is analysed, and national regulations)
- identification of carrier status, which could have reproductive implications or implications for family members
- the test may not find mutations that explain the phenotype
- whether the DNA will be re-tested as new genes are published
- whether residual DNA in the laboratory could be used as a control
- how long the DNA sample will be stored for

Finally, it may be useful to begin a discussion with patients at the stage of consent regarding the concept of 'variants of uncertain significance' and that they may be asked to provide further samples to help validate some of the findings. In addition they can be made aware that other family members may be investigated to clarify cosegregation of a variant of interest.

Laboratory characteristics

The laboratory must be involved in an accreditation scheme (eg, Clinical Pathology Association [CPA], International Organisation for Standardisation [ISO], United Kingdom Accreditation Service [UKAS]) and must participate in external quality assessment (EQA) for NGS. Where none exist, the laboratory should make a reasonable effort to carry out some sample sharing with other similar laboratories to ensure there is an element of external assessment. Where the laboratory carrying out the NGS does so on DNA samples it receives from external laboratories, these laboratories should also be accredited.

Pretesting

Clinical-grade NGS will ensure that variants are reported against the phenotype of the patient and where this is unavailable, the laboratory should make reasonable attempts to obtain it from clinicians or issue a limited report if there is no phenotypic information.

Assessing pathogenicity

Assessing pathogenicity in the laboratory according to the ACMG guidelines as stated above is critical as numerous publications assigning pathogenicity in the literature predate the ACGM recommendations. In some cases, the variants described are accompanied by sufficient functional evidence to confirm pathogenicity. As this is not universally the case, it is our recommendation that original publications are reviewed to assess this.

Panel characteristics

All clinically relevant regions of known genes should be included and at least 90% of the relevant/coding regions must be covered by the sequencing. The coverage at individual genes may be included in the report, or merely retained and stored in the laboratory and available upon request. Sufficient sequencing into the introns is needed to identify splice variants and should cover ±50 bp to ensure branch point mutations are covered. All types of known causative variants should be detected by the method—for example this would include variants localised in the promoter region (eg, 5′-aminolevulinate synthase 2 [*ALAS2*] GATA1 binding site in promoter[23] or for PKLR).[23]

Depth of coverage should be sufficient to detect a heterozygous variant, including single-nucleotide variants, indels and structural variants including CNVs.

Very common pathogenic variants should be reliably detected. Panels have to be large enough to detect phenotypic overlap.[1,6] In cases of syndromes, where anaemia may be the presenting feature, these genes can also be included in the panel (eg, TRNA nucleotidyl transferase 1 [*TNRT1*], carbamoyl-phosphate synthetase 2, aspartate transcarbamylase, and dihydroorotase [*CAD*] or ATP-binding cassette subfamily B member 7 [*ABCB7*]). Genes that are not known to be pathogenic should not be included unless specific research consent has been sought.[24]

Validation of the test should be carried out

As with the introduction of any new test, robust validation should be carried out before introduction into the diagnostic laboratory. Validation should be based on the Rehm 2013 criteria and are summarised in Suppl. Criteria Summary. Depending on the country, some laboratories only need to validate the technology, while others need to validate each individual panel.

Quality control metrics in the report

The report should provide also the following data: average sequencing depth, coverage (target regions)_20×, coverage (target regions)_50×.

Filtering

Filtering is generally set to exclude variants occurring at an allele frequency of >1% in the general population, although this may vary depending on the patient's ethnic group, which should ideally be available for each patient. Any variant with a ClinVar annotation of pathogenicity should be included irrespective of the allele frequency. In HS, common pathogenic modifiers are present above the 1% cut-off (eg, alpha-LeLY/LePRA) and should be specifically sought in patients where HS/hereditary elliptocytosis is a possible diagnosis, especially in the presence of other spectrin alpha, erythrocytic 1 (*SPTA1*) variants. It should be noted that while alpha-LeLY is usually included in most t-NGS panels, alpha-LePRA (c.4339-99C>T) is not usually covered with standard t-NGS platforms. However, alpha-LePRA is frequently co-inherited with the α-Bughill allele[25] (missense variant p.[Ala970Asp]) that is covered by standard t-NGS platforms.

Confirmation of variants

Sanger sequencing is not needed routinely to confirm individual variants if the panel has been validated and the whole pathway is assessed as part of accreditation. However, in some cases Sanger sequencing is necessary before a report can be issued. This includes any unusual variant allele frequency (VAF) or low-coverage reads (eg, 30 reads); any novel CNVs should be confirmed where possible, as should any variants that are present in only forward or reverse reads, as this may result from allele dropout and erroneously suggest homozygosity. Sanger sequencing can also be used to confirm that the correct sample has been tested and therefore the report is being issued to the recipient intended. Confirmation of patient identity may alternatively be provided by DNA analysis of a new sample or analysis of polymorphisms (snapshot technique).

Phenotypic/pedigree confirmation

Pedigree confirmation or additional functional tests may increase or decrease pathogenicity of individual variants as per the ACMG

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guidelines. It is best practice to request samples from relevant family members upfront so that family studies may be carried out, including testing for de novo inheritance. When a clearly pathogenic variant is identified in only one allele of a recessive gene that is associated with the clinical phenotype, it is common practice to report that the pathogenic variant may be *contributing* to the phenotype. The report should be clear on the limitations of the technique and further work can be suggested where this would be helpful.

Assessment for false-negative rate

The genes that have been assessed, the coverage of those genes and the types of genetic abnormalities must be stated. This is to ensure that if a report states that no pathogenic variants were detected, the clinician is clear about what possible causes of the condition have not been excluded. This includes common pathogenic variants not detected by the current method. Other ways of picking up deletions include, for example, MLPA, array comparative genomic hybridisation and single-nucleotide polymorphism arrays. The false-negative rate will be different for each phenotype that is being assessed by the panel, depending on what proportion of the phenotype is known to be explained at the molecular level. This is likely to improve in time as novel genes are identified.

Report writing

The clinical report should be easy to read and be understood by clinicians. The technical characteristics of the test should be made available upon request. Details regarding which transcript was used should be clear.

Recommendations

- Consent for DNA analysis should be obtained (IIB)
- Phenotypic information, and functional test results, if available, should be provided and the genetic variants assessed against this (IIB)
- It should be clear which genes have been tested, the coverage of these genes, and whether or not CNVs are detectable by this method (IIB)
- Filtering of variants against general population should be set between 1% and 5% (IIB)
- Variants with a ClinVar entry and with known modifier effect (eg, alpha-LeLY and LePRA) should be sought and included in the report (IIB)
- Ideally a false-negative rate should be provided, and it should be made clear that a lack of genetic diagnosis does not exclude an appropriate clinical diagnosis (IC)
- Class 4 and 5 variants should be reported if they correspond to the phenotype. In occasional circumstances, some class 3 variants may be reported with careful wording of why they were included (IC)
- The whole panel must be fully validated before use (IA)

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REVIEW PROCESS

Members of the writing group will inform the writing group Chair, Noémi Roy, if any new pertinent evidence becomes available that would alter the strength of the recommendations made in this document or render it obsolete. The document will be archived and removed from the BSH/EHA current guidelines website if it becomes obsolete. If new recommendations are made an addendum will be published on the BSH and EHA guidelines websites.

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